

TestPilot AI - Intelligent Test Generation Framework

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Description

A pip-installable test generation framework using AST parsing. Generates comprehensive pytest suites with boundary value analysis, equivalence partitioning, and LLM-enhanced generation.

Tech Stack

click, pydantic, pytest, jinja2

Features

- AST-based code analysis
- Boundary value test generation
- Equivalence partitioning
- LLM-enhanced generation (Claude CLI default)
- pytest plugin
- Coverage gap detection

Sample Input

Python source files
`testpilot generate --source src/ --output tests/`

Sample Output

Generated pytest test files
`testpilot analyze --source calculator.py`

Project Files

```
.gitignore
LICENSE
project-idea.json
pyproject.toml
README.md
.github/workflows/test.yml
samples/calculator.py
samples/string_utils.py
src/testpilot/analyzer.py
src/testpilot/cli.py
```

```
src/testpilot/generator.py
src/testpilot/models.py
src/testpilot/plugin.py
src/testpilot/py.typed
src/testpilot/report.py
src/testpilot/strategies.py
src/testpilot/__init__.py
src/testpilot/__version__.py
src/testpilot/providers/api_provider.py
src/testpilot/providers/base.py
src/testpilot/providers/claude_cli.py
src/testpilot/providers/none_provider.py
src/testpilot/providers/__init__.py
src/testpilot/templates/test_module.py.j2
src/testpilot_ai.egg-info/dependency_links.txt
src/testpilot_ai.egg-info/entry_points.txt
src/testpilot_ai.egg-info/PKG-INFO
src/testpilot_ai.egg-info/requires.txt
src/testpilot_ai.egg-info/SOURCES.txt
src/testpilot_ai.egg-info/top_level.txt
tests/conftest.py
tests/test_analyzer.py
tests/test_cli.py
tests/test_generator.py
tests/test_models.py
tests/test_plugin.py
tests/test_providers.py
tests/test_report.py
tests/test_strategies.py
tests/__init__.py
```

Test Results

PASSED

Run: 66 | Passed: 66 | Failed: 0